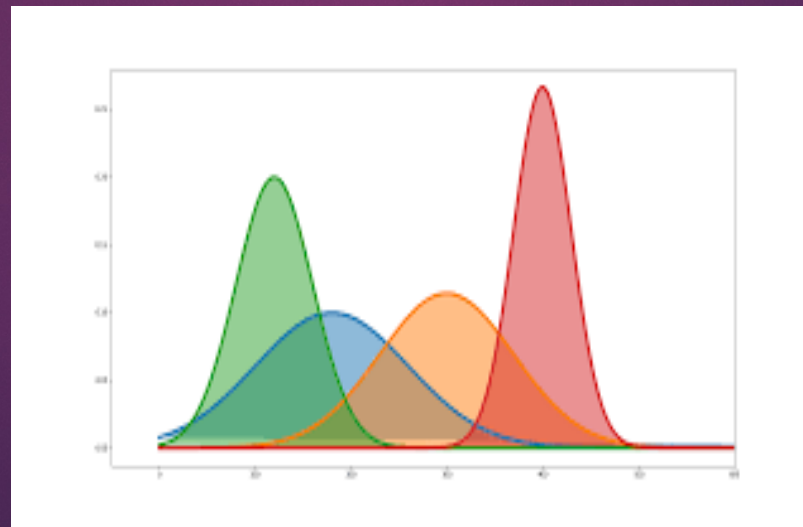


Seminar: Statistical Distributions



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- Hence, we have **two possible** values for a Bernoulli distribution: **1, with probability of p, and 0, with probability 1-p.**
- Then **mean** of a Bernoulli distribution with probability p is p: **$E(x) = p$**
- The **variance** is: **$\text{var}(x) = p(1-p)$** and the standard deviation is $\sigma = \sqrt{p(1-p)}$

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For example, $X = 1$ if an individual from the dataset is student, and 0 otherwise. If we have 1000 observations and we know that 250 of them are students, then what is the probability of an individual to be student? What is the mean of X?

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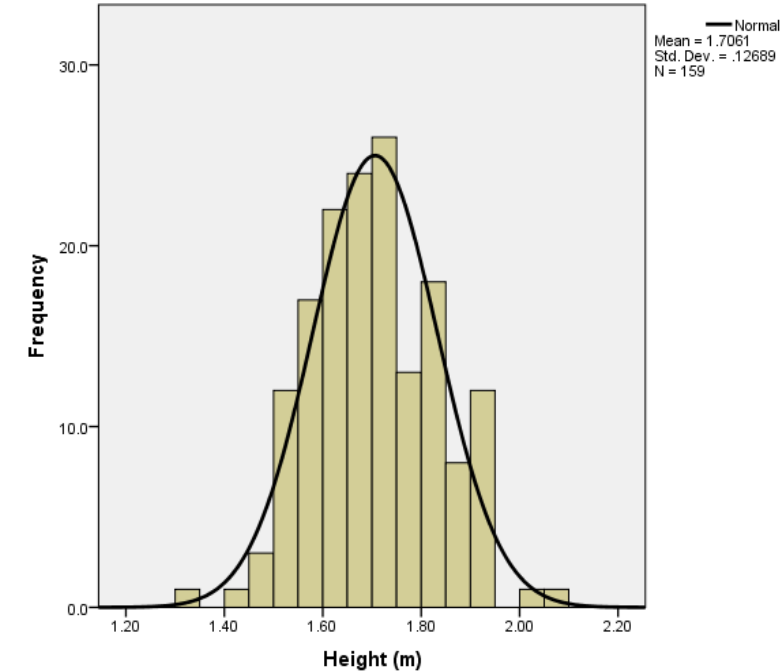
$$p = 250/1000 = 0.25 \text{ and } E(x) = p = 0.25$$

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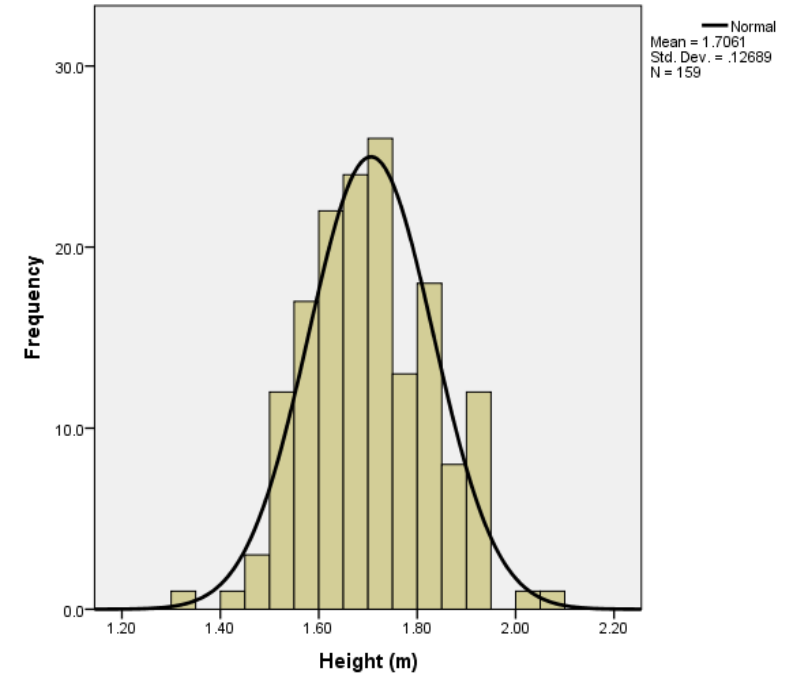


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Normal distribution: used to model continuous data, and this distribution is the main idea of statistical distributions.

- For the normal distribution we need the **mean μ** and the **std. deviation σ**
- the area under the curve is 1
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Rather than asking: “What’s the probability someone is precisely 1.84m tall?” (to which the answer is 0), we ask: “What’s the probability someone is at least 1.84m tall?”



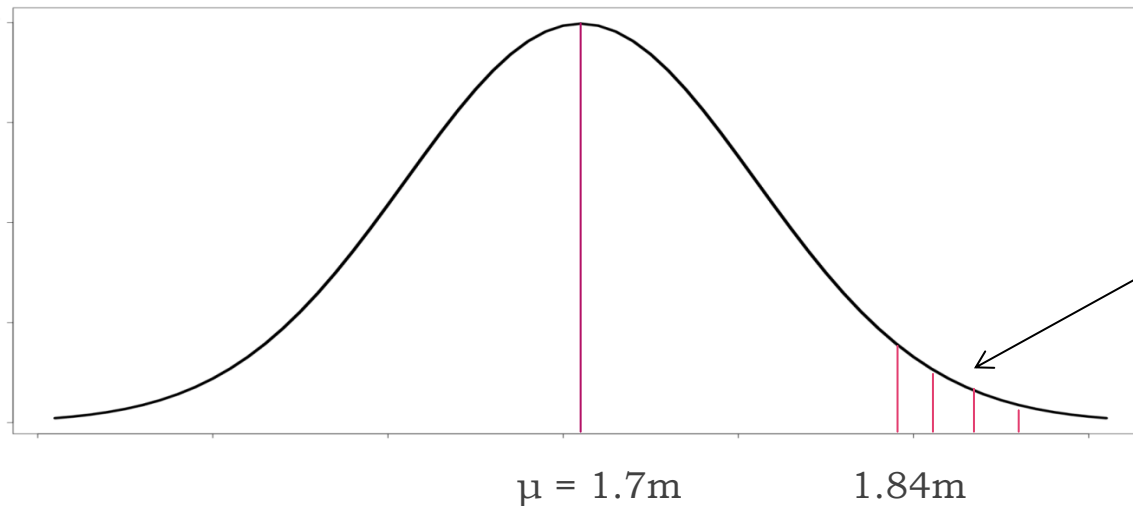
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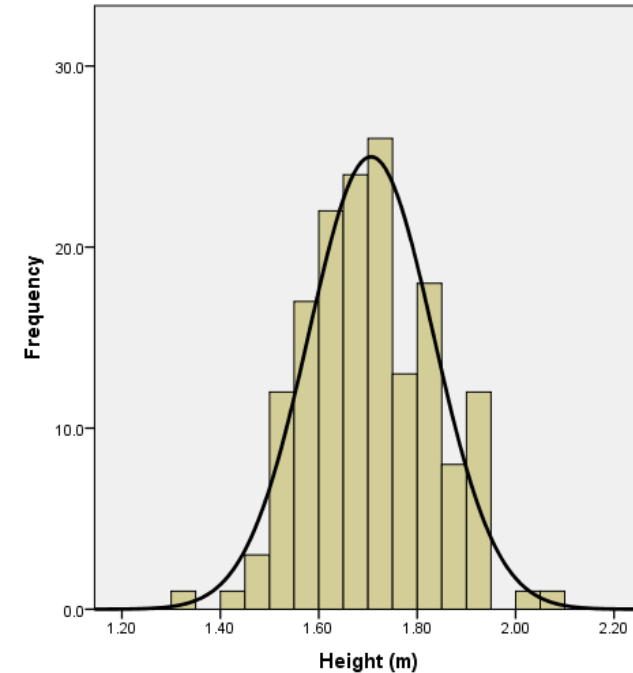
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area we want (prob of being above 1.84m)

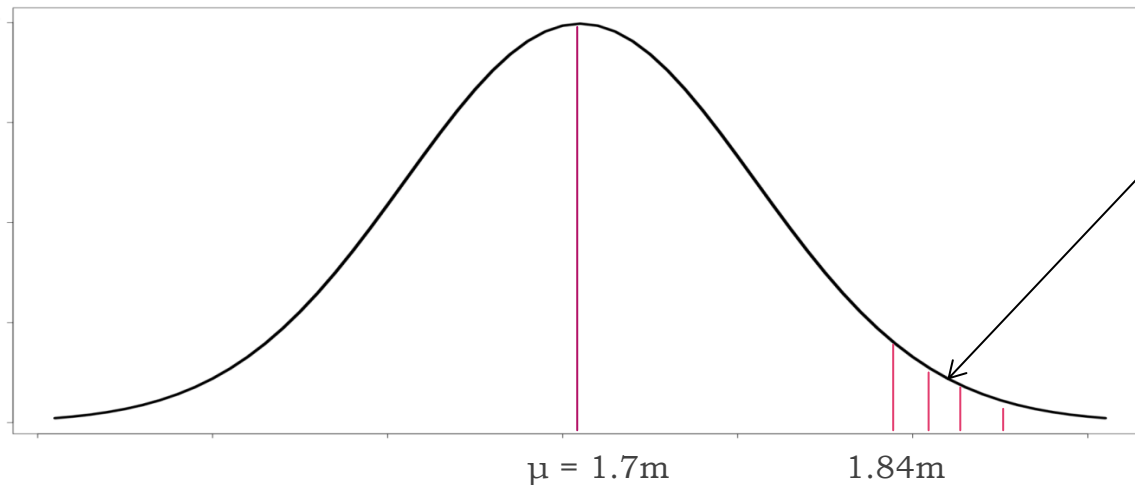


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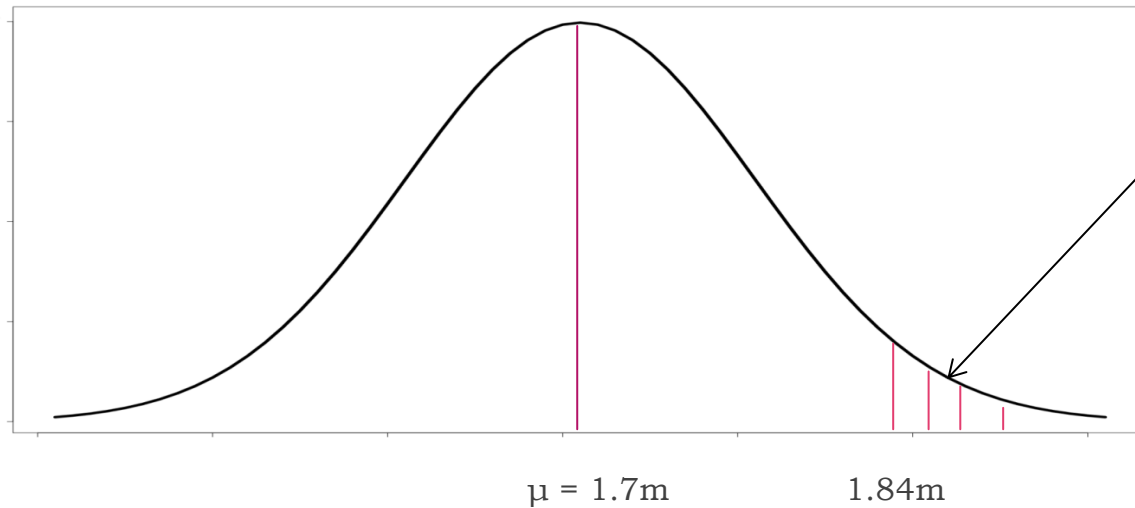
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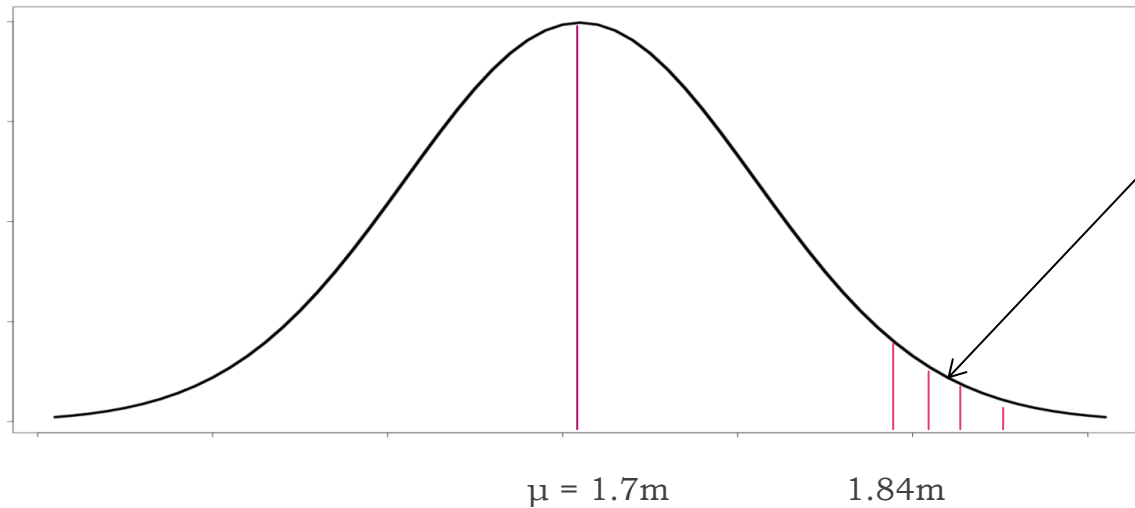
$$\text{step 2. } z = \frac{x - \mu}{\sigma} = \frac{1.84m - 1.71m}{0.13m} = 1.00$$

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Step 3. look at the standard normal table (see vle handout) on the side that has 0.00 as the top entry at the left. Then, look down the left column until you see 1.00 and take the first column (0.00). You should see **0.84134**; this is the entry for 1.00. Thus, the area on the right is $1 - 0.84134 = 0.158$. Hence, the probability of someone being above 1.84m tall is about 0.16 or 16%.

Recall from lectures...

Normal distribution

Remember, the standardisation process:

step 1. $Z = \frac{x-\mu}{\sigma}$ and step 2. $z = \frac{x-\mu}{\sigma} = \frac{1.84m-1.71m}{0.13m} = 1.00$ which yields from the table 0.84134

ALWAYS: the left-hand column is for the first two digits of z, and the top-most row is to specify the third digit.

Examples:

What is the value for $z = 1.23$?

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What is the value for $z = 1.23$? A value for $z = 1.23$ would be found on row 1.20 and column 0.03. That is 0.89065

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That is $p = 29/57 = 0.508$

Then we calculate the variance, which is $p(1-p) = 0.249$

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$$\mu = p = 33/61 = 0.54$$

$$\sigma^2 = p(1-p) = 0.25$$

Question 3

In a group of 7 items, each of which has a probability of 0.6 of being of interest, what's the probability precisely 3 are of interest?

Hint! This is a binomial distribution question. The number of items is n , the probability given is p , and the number needed of interest is r . Use the Binomial probability formula to calculate $P(r)$.

Remember: $P(r \text{ successes in } n \text{ goes}) = C \times p^r \times (1 - p)^{n-r}$, where $C = \frac{n!}{r!(n-r)!}$, $p = 0.6$, $n = 7$ and $r = 3$. So....

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$$C = \frac{7!}{3!(7-3)!} = 35$$

$$\Pr(r=3) = 35 \times 0.6^3 \times (1 - 0.6)^{7-3} = 0.1935$$

Question 4

A box contains 125 widgets, and the manufacturing process is known to produce defective widgets 2.4% of the time. Use the binomial distribution to calculate the probability of precisely the expected number of defective widgets in a box.

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$$C = \frac{125!}{3!(125-3)!} = 317,992.59$$

$$\Pr(r=3) = C \times 0.024^3 \times (1 - 0.024)^{125-3} = 0.227$$

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$$\text{Hence } \sigma^2 = 38 * 0.54 * (1 - 0.54) = 9.44$$

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Hence, $z = (7 - 97) / 7.47 = -12.05$

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Answer:

Then $z * \sigma = x - \mu \Rightarrow x = z * \sigma + \mu = 73.37$

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Items are made with weights following a normal distribution with mean 31.16g and standard deviation 3.05g. What's the probability an item is made with a weight below 29.70g?

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1st step: standardise: $Z = \frac{x-\mu}{\sigma} = \frac{29.7-31.16}{3.05} = -0.4786$

2nd step: look at the table to find the value:

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Answer: 0.31561

Question 9

NutriBox makes individual servings of healthy snack foods. Their “Berry Bonanza” box is filled with a random assortment of dried berries, and weighs 43g on average, with a standard deviation of 1.4g. Any box of Berry Bonanza over 45g gets sent to the premium NutriBox customers. What proportion of boxes is this?

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1st step: standardise: $z = \frac{x - \mu}{\sigma} = \frac{45 - 43}{1.4} = 1.428$

2nd step: look at the table to find the value: 0.92364

Answer: $1 - 0.92364 = 0.0763$

Question 10

If a distribution has mean 55 and standard deviation 6.45, what is the lower bound of the 95% prediction interval?

Remember, the 95% prediction interval is the interval in the data that contains 95% of the distribution

Use this to find z .

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If a distribution has mean 55 and standard deviation 6.45, what is the lower bound of the 95% prediction interval?

Remember, the 95% prediction interval is the interval in the data that contains 95% of the distribution

Use this to find z. To find the lower bound, we need to find the value for 2.5%.

$$Z = -1.96$$

Remember: $z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z * \sigma = 55 - 1.96 * 6.45 = 42.36$

Sum up...

1. Bernoulli Distribution: mean and variance
2. Binomial Distribution: probability formula to calculate $P(r)$
3. Normal Distribution: mean and variance, standardisation, predictions intervals